RIWA KARAM

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Irvine, California

Objective

Highly motivated electrical and computer engineer, with a vast imagination and a passion for robotics and machine learning. Great teamwork and communication skills allow me to work well in groups. Open to opportunities that will help me grow in my field and expand my knowledge and my skill set, while enjoying the work and fueling my curiosity.

Industry Experience

- **ASML** [] *June 2025 – September 2025*
  Mechatronics Research Engineer Intern Wilton, CT
  - Investigated machine health monitoring using statistical modeling and Bayesian hierarchical methods for the purpose of uncertainty quantification and structural modeling.
  - Developed a Bayesian hierarchical model using Python and pymc library to predict drift effects and performance degradation, enabling root cause analysis across machine lifetimes.
  - Developed other machine learning models such as decision tree, random forest, multiple regression, and other toy regression models on subsets of the data to compare results with the Bayesian model and understand the data better.
  - Conducted data collection, data analysis, and data pre-processing using Python and MATLAB including regression modeling, principal component analysis (PCA), and heatmap-based pattern recognition.
  - Learned and applied methods like Monte Carlo Markov Chains (MCMC), Bayesian statistics, statistical methods, and time-lag regression techniques.
- **ASML** [] *June 2024 – September 2024*
  Mechatronics and Control Systems Engineer Intern Wilton, CT
  - Developed a Gaussian Process Regression (GPR) model using Python and MATLAB to predict the force generated by a piezo-electric actuator having up to ±2 Newtons of accuracy, which led to an ASML invention.
  - Conducted data analysis and multiple toy regressions and machine learning models to understand the correlations and co-dependencies in the data.
  - Automated and merged MATLAB and Python scripts that run testing workflows for both the mechatronics and software teams, reducing runtime from 40 hours to an overnight run.
  - Investigated feasibility of optimization problems and current-control hardware issues for test stands by analyzing their electrical circuitry and alternative hardware.
- **AviaPro Consulting Inc.** [] *June 2022 – July 2023*
  Software Developer Remote
  - Developed several web applications using React.js and Node.js, migrating the company from excel-based repositories to a streamlined interface with multiple interconnected databases, in addition to other internal tools.
  - Applied software DevOps practices such as Continuous Integration / Continuous Development (CI/CD), Infrastructure as a Code, and Git.
  - Contributed to multiple full-stack web development projects as well as a solo full-stack web dev project.
  - Started as a summer co-op (June 2022 - August 2022) and then transitioned to part-time starting September 2022.

Academic Experience

- **University of California, Irvine** [] *September 2023 - Present*
  Graduate Student Researcher Irvine, CA
  - Conducting research in collaborative multi-agent systems, optimization, machine learning, ecology robotics, coverage control, formation control, and human-swarm interaction, under the supervision of Professor Magnus Egerstedt at the Robot Ecology Lab and Professor Yanning Shen.
  - Actively helping to maintain the Robot Ecology Lab Robotarium testbed, which consists of testing, debugging, and fixing robots, as well as other troubleshooting processes.
  - Presented my first author accepted regular paper at the 64th IEEE Conference on Decision and Control (CDC) in Rio de Janeiro, Brazil.
  - Presented research findings in person at the 45th SoCal Control Workshop at the University of California, San Diego.
  - Presented a poster on my master thesis at the 2025 UCI GEECS Annual Technology Showcase.
  - Mentored undergraduate student lab members and worked together on research projects.
  - Co-wrote the Robot Ecology Lab manual, which consists of all in-house debugging processes, tutorials on how to run experiments in the testbed, and Vicon system maintenance.
  - Conducted multiple lab tours with other lab members to more than 100+ visitors, which included live demonstrations of multiple homogeneous robots in the Robotarium performing collaborative algorithms.

Education

- **University of California, Irvine** [🌐] *September 2023 - Expected June 2028*
  Electrical & Computer Engineering, Doctor of Philosophy Irvine, CA
  - Robot Ecology Lab
  - Advisors: Professor Magnus Egerstedt, Professor Yanning Shen
- **University of California, Irvine** [🌐] *September 2023 - June 2025*
  Electrical & Computer Engineering, Master of Science Irvine, CA
  - GPA: 3.975/4.00
  - Thesis: A Graphical Interface for Specifying and Establishing Multi-Robot Formations
  - Relevant Coursework: Optimization, Network Science, Control and Machine Learning, Quantum Computing
- **University of Balamand, Al-Kurah** [🌐] *August 2020 - June 2023*
  Computer Engineering, Bachelor of Science Balamand, Al-Kurah, Lebanon
  - GPA: 3.90/4.00
  - Computer Engineering Class of 2020 Valedictorian
  - Graduation Project: Gamification of Virtual Reality Kitchen Scenarios
  - Tutored peers through group and private sessions in courses such as programming (Python, MATLAB, C), Signals and Systems, Microcontrollers, Embedded Systems, and Computer Architecture.

Publications

C=Conference, T=Thesis, S=Submitted

[S.1] Riwa Karam, Alexander A. Nguyen, Ruoyu Lin, David R. Martin, Diana Morales, Brooks A. Butler, and Magnus Egerstedt, **Collaboration in Multi-Robot Systems: Taxonomy and Survey over Frameworks for Collaboration.** arXiv preprint, DOI.

[C.1] Riwa Karam, Ruoyu Lin, Brooks A. Butler, and Magnus Egerstedt, **Resource Allocation for Multi-Team Collaboration Based on Hamilton's Rule**, IEEE 64th Conference on Decision and Control (CDC), Rio de Janeiro, Brazil, 2025, pp. 6891-6898, DOI

[T.1] Riwa Karam, **A Graphical Interface for Specifying and Establishing Multi-Robot Formations**, MS Thesis, University of California, Irvine, 2025, DOI

Skills

- **Programming Languages and Libraries:** MATLAB, Python, Scikit-learn, PyTorch, Keras, NumPy, Pandas, Matplotlib, Shell Scripting, pymc, C, Java, C#.
- **Hardware Technologies:** Vicon Motion Capture (MoCap) System, Robotarium, robot building and debugging.
- **Other Tools & Technologies:** Git & GitHub, LateX, Simulink, VS Code, Unity, Microsoft Tools, Google Tools
- **Soft Skills:** Research, Time Management, Written and Verbal Communication, Critical Thinking

Honors, Awards, and Certifications

- **Faculty Award for Academic Excellence** *July 2023*
  University of Balamand, Faculty of Engineering [🌐]
- **First Place Winner (1st out of 12 teams), Lebanon IoT & AI Challenge 2021** *November 2021*
  Arab IoT & AI Challenge [🌐]
- **Dean's Honor List** *August 2020 - June 2023*
  University of Balamand, Faculty of Engineering [🌐]

Languages

- Arabic: Native Language, proper Arabic and Lebanese Arabic
- English: High Proficiency, language of education in college
- French: High Proficiency, language of education in school
- Spanish: Beginner, Duolingo Score of 38 (A2 level of CEFR)

Extra-Curricular Activities

- **Sports played:**
  - High School Volleyball Varsity Team Captain (opposite hitter, middle blocker, outside hitter)
  - University Volleyball Varsity Team (outside hitter)
  - Middle School Ping-Pong Varsity Team
  - Club Gymnastics from age 5 to 14
- **Musical Education:** Violin & Music Theory, Lebanese National Higher Conservatory of Music, age 8 to 14
- **Other Interests:** biking, surfing, swimming, board games, reading books, watching movies

References available upon request